

THE MICROPROCESSORS & MICROCONTROLLERS

Instructor: MSc. The Tung Than

PRACTICE EXERCISE #6:

IO PROCESSING, CALCULATION AND MEMORY ON THE 8086 MICROPROCESSOR

I. Student preparation

- Students install the [emu8086 software](#)
- Students understand how to access and use memory on 8086 microprocessor
- Students understand the assembly language instruction set on the 8086

II. Practice content (5 points)

1. Refer to the example:
 - Students open File >> example >> add/subtract
 - Run test, based on the instruction set describing its operation.
 - (Help >> Documentations and tutorials or Press F1)
2. Write a program that adds/subtracts 2 32-bit numbers.

III. Exercise (5 points)

1. Enter a 2-digit number N from the keyboard through the console screen.
2. Print the first N Fibonacci numbers to the screen.

IV. Report

Compress design files and report files into a file named as follows:

[<LAB...>]-[<Student code>]-Full name

The required report file contains the following contents:

1. Flowchart of the above request processing algorithm.
2. Explain how the algorithm works, accompanied by a video (send a Google Drive link) to demonstrate the result in case the instructor cannot run the design file.